Epidemiology of the 2026 Measles Outbreak in Bangladesh: A Descriptive Analysis of National Surveillance Data with Vaccine Coverage Assessment

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Data & Code: https://github.com/khalilurrrahmanridoykhan/bangladesh-measles-2026

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Abstract

Background: Bangladesh has experienced a large-scale measles outbreak since March 2026, coinciding with a national Measles-Rubella (MR) immunisation campaign. This study describes the epidemiological characteristics of the outbreak using publicly available daily surveillance data published by the Directorate General of Health Services (DGHS).

Methods: We extracted structured data from 62 consecutive daily press release PDFs published by DGHS (2 April to 2 June 2026) using automated web scraping and PDF text extraction. We analysed temporal trends, geographic distribution across eight administrative divisions, mortality, and the relationship between MR campaign vaccination coverage and division-level incidence rates.

Results: A total of 73,362 suspected and 9,136 confirmed measles cases were reported (confirmation rate 12.5%). A total of 59,106 patients were hospitalised and 54,812 (92.7%) recovered. Suspected and confirmed deaths numbered 504 and 90, respectively (confirmed case fatality rate 0.99%). The outbreak grew rapidly with a doubling time of 6.1 days. Peak daily suspected cases reached 1,503 on 10 May 2026. Dhaka division accounted for 47.0% of all confirmed cases (incidence 95.7 per 100,000 population). National MR campaign coverage was 102%; however, Pearson correlation between division-level coverage and incidence was non-significant ($r = 0.10$, $p = 0.82$).

Conclusions: This outbreak represents one of the largest measles events recorded in Bangladesh. Despite high MR campaign coverage, urban transmission persisted - particularly in Dhaka. Strengthening routine immunisation, improving cold-chain management, and targeting high-density urban populations are essential to interrupt transmission.

1. Introduction

Measles remains one of the most contagious infectious diseases known to medicine, with a basic reproduction number (R0) estimated between 12 and 18. Despite the existence of a safe, effective, and affordable vaccine, measles caused an estimated 136,000 deaths globally in 2022, the majority among children under five years of age (World Health Organization [WHO], 2023). Following disruptions to routine immunisation services during the COVID-19 pandemic (2020-2022), a global resurgence of measles has been observed, with large outbreaks reported across Africa, South-East Asia, and the Eastern Mediterranean region.

Bangladesh, a densely populated lower-middle-income country with approximately 170 million inhabitants, has maintained an Expanded Programme on Immunisation (EPI) since 1979. The measles vaccine was introduced into the national schedule in 1980, and a two-dose schedule - measles-rubella (MR) vaccine at 9 and 15 months - has been in place since 2012. Bangladesh had previously achieved measles vaccination coverage exceeding 95% in many districts, and no major nationwide outbreak had been documented for several years prior to 2026.

In March 2026, the DGHS activated its Integrated Control Centre in response to a sharp rise in febrile rash illness consistent with measles across multiple divisions. Daily situation reports were published publicly from 15 March 2026, providing a rare opportunity for granular, real-time epidemiological analysis using government surveillance data. By 2 June 2026, more than 73,000 suspected cases had been reported nationally.

This study has three objectives: (i) to describe the temporal trajectory of the 2026 Bangladesh measles outbreak using 62 days of daily national surveillance data; (ii) to characterise the geographic distribution of the outbreak burden across Bangladesh's eight administrative divisions; and (iii) to assess the relationship between the 2026 national MR immunisation campaign coverage and division-level measles incidence.

2. Methods

2.1 Data Source

Data were sourced exclusively from publicly available daily press release PDFs published by the DGHS of the Government of Bangladesh (https://dghs.gov.bd/pages/press-releases/). Each press release is a structured three-page PDF document reporting: (i) national suspected and confirmed measles cases in the preceding 24 hours and cumulatively from 15 March 2026; (ii) suspected and confirmed deaths; (iii) hospitalisation and recovery counts; (iv) division-level breakdowns; and (v) Measles-Rubella Campaign 2026 vaccination coverage by division.

2.2 Data Collection

Automated data collection was conducted using custom Python scripts (Python 3.14; libraries: requests, BeautifulSoup4, pdfplumber). Bangla numeral characters were converted to Western Arabic numerals prior to parsing. All scripts and clean datasets are available at: https://github.com/khalilurrrahmanridoykhan/bangladesh-measles-2026.

2.3 Case Definitions

A suspected case is any person with fever and generalised maculopapular rash plus at least one of: cough, coryza, or conjunctivitis. A confirmed case is a suspected case with laboratory confirmation (measles IgM serology or viral isolation) or epidemiologic linkage to a laboratory-confirmed case (WHO 2018 surveillance guidelines).

2.4 Statistical Analysis

Descriptive statistics were computed for national and division-level indicators. Doubling time during the exponential growth phase (first 14 days) was estimated using log-linear regression: ln(cumulative suspected cases) = a + b x day, with doubling time = ln(2)/b. Incidence rates were calculated per 100,000 population using 2022 Bangladesh Census denominators. Pearson correlation assessed the relationship between division-level MR campaign coverage and confirmed incidence. All analyses used Python 3.14 (pandas, matplotlib, scipy). Significance: $p < 0.05$.

2.5 Ethics

This study used only publicly available, aggregated government surveillance data. No individual patient data or identifiable information were accessed. Formal ethics review was not required.

3. Results

3.1 National Overview

Over the 62-day study period (2 April to 2 June 2026), 73,362 suspected and 9,136 confirmed measles cases were reported nationally (confirmation rate 12.5%; Table 1). A total of 59,106 patients were hospitalised, of whom 54,812 (92.7%) recovered. Suspected deaths totalled 504 (suspected CFR 0.69%); confirmed deaths numbered 90 (confirmed CFR 0.99%).

Table 1. National Summary Statistics - Bangladesh Measles Outbreak, 2 April to 2 June 2026

Indicator	Value
Reporting period	2 April 2026 - 2 June 2026 (62 days)
Total suspected cases	73,362
Total confirmed cases	9,136
Confirmation rate	12.5%
Total hospitalised (cumulative)	59,106
Hospitalisation rate (of suspected)	80.6%
Total recovered (cumulative)	54,812
Recovery rate (of hospitalised)	92.7%
Total suspected deaths	504
Total confirmed deaths	90
Case fatality rate - suspected	0.69%
Case fatality rate - confirmed	0.99%
Peak daily suspected cases	1,503 (10 May 2026)
Mean daily suspected cases	1,167

3.2 Temporal Trends

The epidemic curve demonstrated rapid early growth followed by a sustained plateau (Figure 1). Daily suspected cases rose from 685 on 2 April to a peak of 1,503 on 10 May 2026. The mean daily suspected case count was 1,167 (range: 674-1,503). During the final week, daily cases declined below 1,100, suggesting early plateau. The doubling time during the first 14 days was 6.1 days (daily growth rate 11.3%; r2 = 0.94). Week-over-week growth was highest in week 2 (+12.1%), decelerating from week 7 onward (-23.6% in the final partial week).

3.3 Geographic Distribution

Measles burden was highly concentrated in Dhaka division: 34,497 confirmed cases (47.0% of national total; incidence 95.7 per 100,000; Table 2). Chattogram ranked second (12,009 cases; 42.3/100,000). Rangpur had the lowest burden (3,323 cases; 21.2/100,000). Barishal recorded the highest incidence rate (79.7/100,000) despite the smallest population.

Table 2. Division-level Confirmed Measles Cases and Incidence Rates, Bangladesh 2026

Division	Population (2022)	Confirmed Cases	Incidence/100k	% National

Dhaka	36,054,418	34,497	95.7	47.0%
Chattogram	28,423,019	12,009	42.3	16.0%
Rajshahi	18,484,858	9,414	50.9	12.6%
Barishal	8,325,666	6,634	79.7	8.8%
Khulna	15,563,000	5,422	34.8	7.2%
Sylhet	10,009,239	3,688	36.8	4.9%
Rangpur	15,665,000	3,323	21.2	4.4%
Mymensingh	11,370,000	N/A	N/A	N/A
NATIONAL TOTAL	143,895,200	74,987*	52.1	100%

** National total includes cases not attributed to specific divisions in daily reports.*

3.4 Mortality

As of 2 June 2026, 504 suspected and 90 confirmed measles deaths had been recorded cumulatively since 15 March 2026. The confirmed CFR of 0.99% is consistent with measles mortality estimates for low- and middle-income settings. Deaths data were reliably extractable for 14 of 62 reporting days (mid-May to June 2026) due to PDF layout variability in earlier reports.

3.5 Vaccination Coverage and Incidence

The national MR Campaign 2026 achieved 102% overall coverage (18,452,281 doses administered; Table 3). Despite this, the Pearson correlation between division-level MR campaign coverage and confirmed measles incidence per 100,000 was $r = 0.10$ ($p = 0.82$) - not statistically significant. This indicates that high campaign coverage, achieved during an active outbreak, did not prevent ongoing transmission at the division level.

Table 3. MR Campaign 2026 Coverage vs. Confirmed Measles Incidence by Division

Pearson r = 0.10, p = 0.82 (not statistically significant)

Division	Coverage (%)	Confirmed Cases	Incidence/100k
Dhaka	103%	34,497	95.7
Chattogram	103%	12,009	42.3
Rajshahi	103%	9,414	50.9
Barishal	101%	6,634	79.7
Khulna	101%	5,422	34.8
Sylhet	99%	3,688	36.8
Rangpur	103%	3,323	21.2
Mymensingh	102%	N/A	N/A

4. Discussion

4.1 Summary of Findings

This study presents the first systematic epidemiological analysis of the 2026 measles outbreak in Bangladesh, using 62 days of publicly available national surveillance data. The outbreak was characterised by rapid exponential growth (doubling time 6.1 days), high hospitalisation burden (80.6%), and strong geographic concentration in Dhaka (47% of national confirmed cases). Despite achieving over 100% national MR campaign coverage, no statistically significant correlation between vaccination coverage and division-level incidence was observed.

4.2 Rapid Transmission and Urban Concentration

The estimated doubling time of 6.1 days is consistent with a high effective reproduction number ($R_e > 1$), indicating ongoing transmission in a susceptible population. The concentration of 47% of cases in Dhaka reflects its extreme population density (approximately 44,000 persons per km2 in the metropolitan area), high household crowding, and large populations of urban migrants with potentially incomplete vaccination histories. Barishal's high incidence rate (79.7/100,000) despite its small population suggests intense localised transmission in a compact area.

4.3 Vaccination Coverage and Pre-existing Immunity Gaps

The absence of a significant inverse correlation between MR campaign coverage and incidence ($r = 0.10$, $p = 0.82$) is a critical finding. The campaign was reactive - conducted during an active outbreak rather than prophylactically. Pre-existing immunity gaps accumulated over years of suboptimal routine immunisation and pandemic-era service disruptions provided the susceptible pool driving transmission. Division-level coverage data may also mask sub-district heterogeneity where pockets of unvaccinated children sustain local transmission chains even where aggregate coverage appears high.

4.4 Mortality

The confirmed CFR of 0.99% is higher than high-income country estimates (<0.1%) but within the range for low- and middle-income settings (0.1%-5.0%). The high hospitalisation rate (80.6%) reflects either clinical severity or health-seeking behaviour that triages severe cases to hospital. Deaths data were incomplete across the study period, which is a significant limitation.

4.5 Limitations

- Aggregated data only: no individual patient data (age, sex, vaccination status, district) were available, preventing stratified analyses.
- Case definition sensitivity: the 12.5% confirmation rate suggests a substantial proportion of suspected cases may be non-measles febrile rash illnesses.
- Incomplete deaths data: mortality reliably extractable for only 14 of 62 days.
- Division-level vaccination data only (n=8): limits statistical power for correlation.
- No pre-outbreak baseline: press releases began 15 March 2026; early dynamics before this date are not captured.

- Single government data source: independent verification was not possible.

5. Conclusions

The 2026 measles outbreak in Bangladesh is one of the largest on record, with over 73,000 suspected and 9,136 confirmed cases in 62 days, 504 suspected deaths, and a rapid early doubling time of 6.1 days. The outbreak was disproportionately concentrated in Dhaka, with no statistically significant relationship between MR campaign coverage and division-level incidence, highlighting the limitations of reactive vaccination during active transmission.

Three urgent policy priorities emerge: (i) strengthening routine two-dose MR immunisation to maintain population immunity above the elimination threshold (>=95% two-dose coverage); (ii) implementing targeted supplementary immunisation in high-density urban communities well in advance of outbreak risk periods; and (iii) improving real-time surveillance infrastructure, including laboratory confirmation capacity, for timely epidemiological response.

Data Availability

All data, analysis scripts, and figures: https://github.com/khalilurrrahmanridoykhan/bangladesh-measles-2026
Primary data source: https://dghs.gov.bd/pages/press-releases/

Competing Interests

The author declares no competing interests.

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Figures

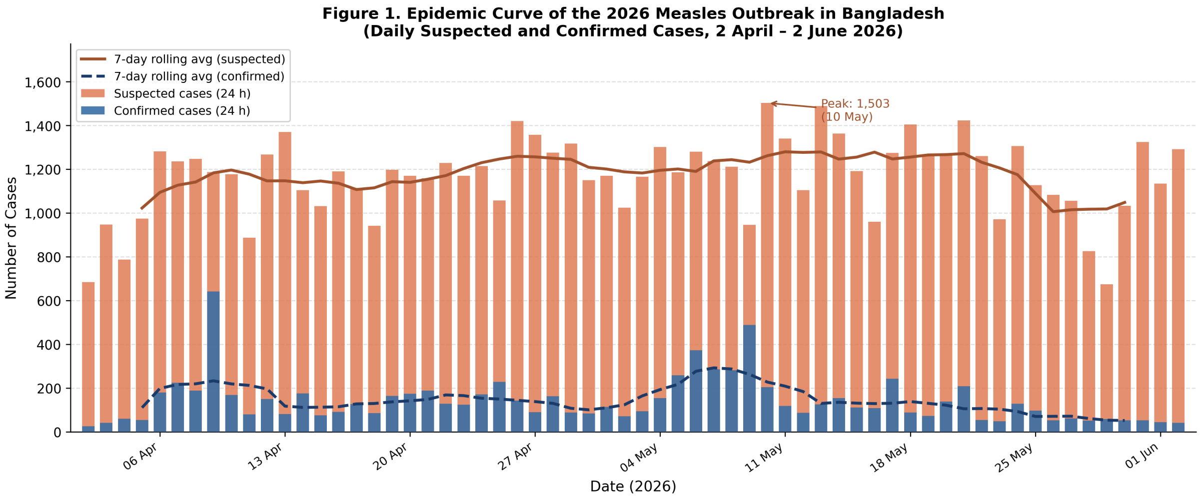

Figure 1. Epidemic curve of the 2026 measles outbreak in Bangladesh. Daily suspected cases (orange) and confirmed cases (blue) with 7-day rolling averages. 2 April to 2 June 2026.

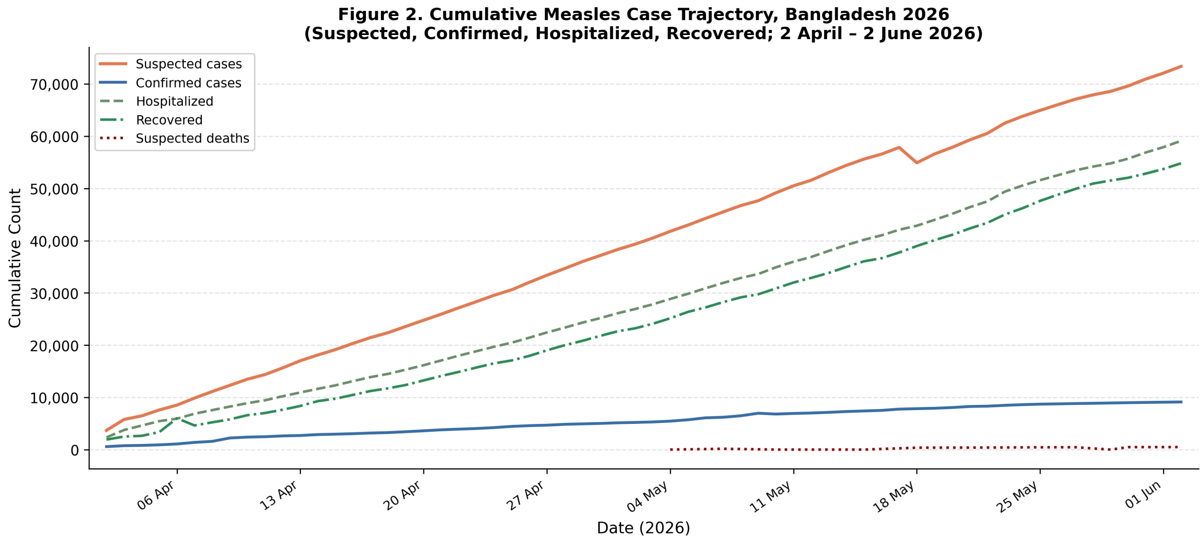

Figure 2. Cumulative trajectory of measles cases, hospitalisations, recoveries, and suspected deaths, Bangladesh 2026.

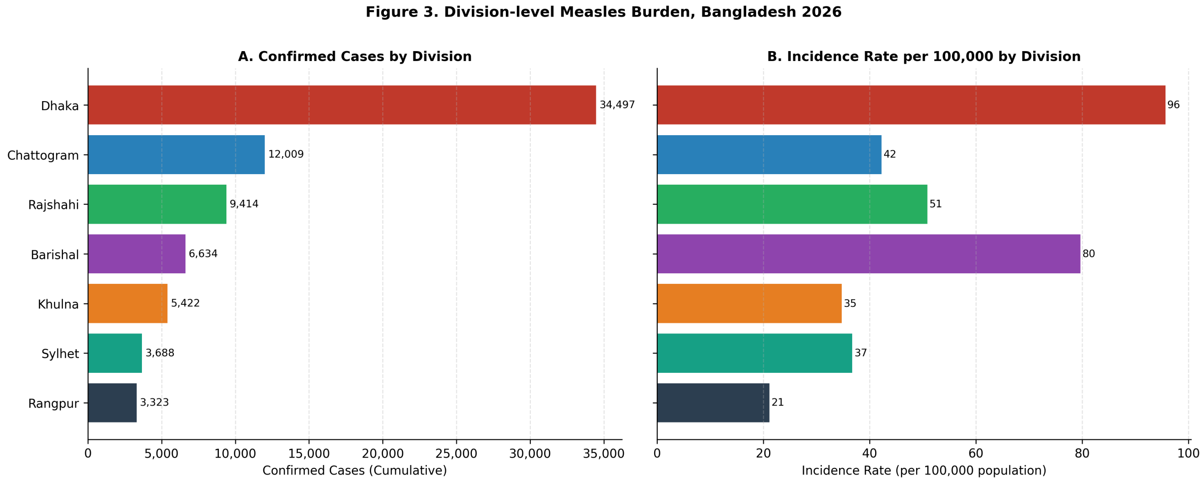

Figure 3. Division-level measles burden, Bangladesh 2026. Panel A: Confirmed cases (cumulative). Panel B: Incidence rate per 100,000 population.

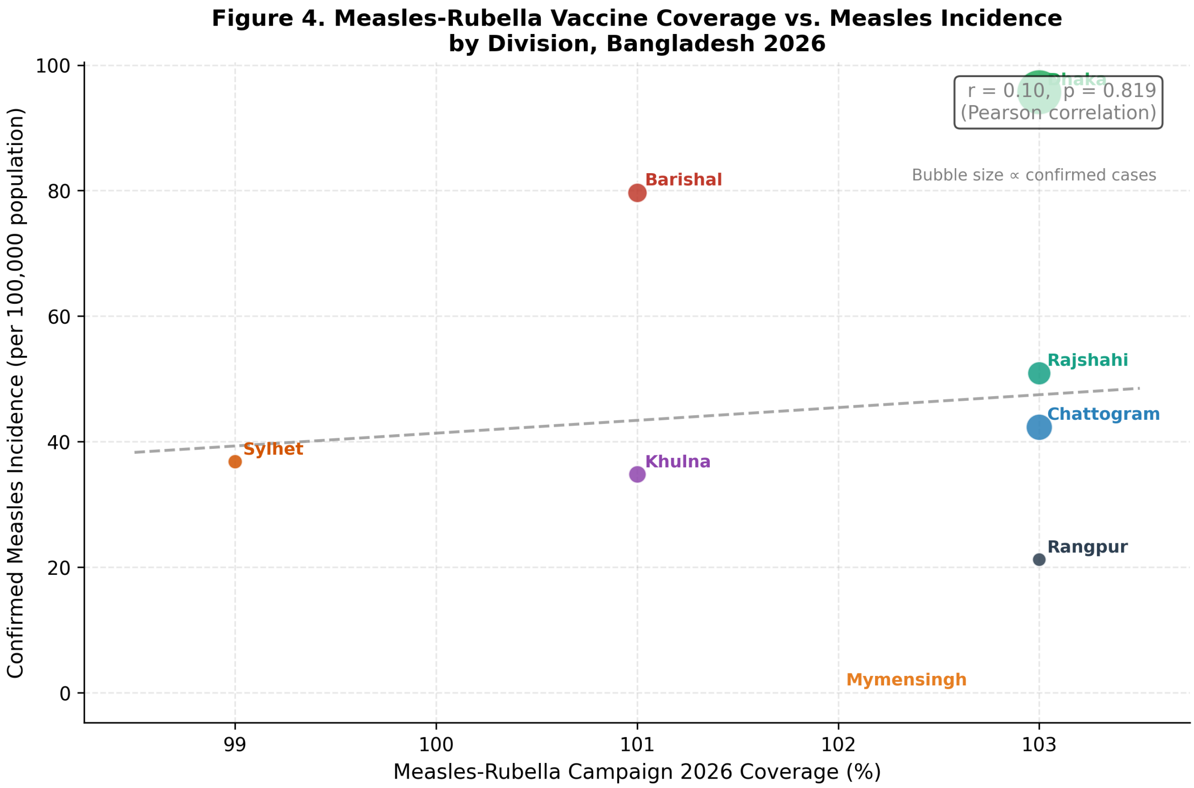

Figure 4. MR Campaign 2026 coverage (%) vs. confirmed measles incidence per 100,000 by division. Bubble size proportional to confirmed cases. Pearson r = 0.10, p = 0.82.